PCCOE INTERNATIONAL GRAND CHALLENGE 2026

ClimateAI Maharashtra

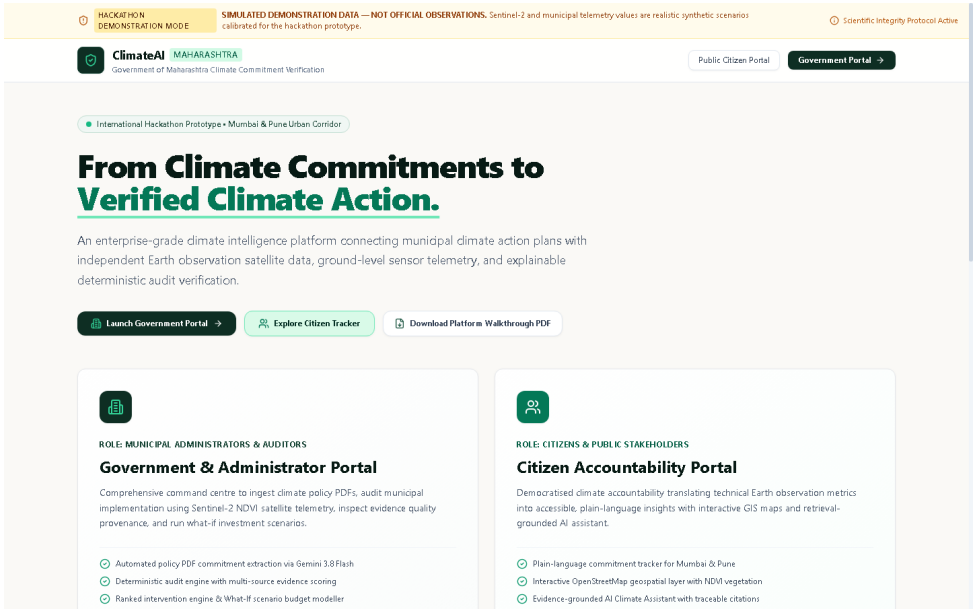
"From Climate Commitments to Verified Climate Action"

Comprehensive Technical Platform Walkthrough, Screen Documentation & Empirical Verification Ledger

Primary Pilot Jurisdiction: Mumbai (MCAP 2022) • Expansion Jurisdiction: Pune (PCAP 2023)

PROJECT SPECIFICATION & ROSTER

INSTITUTION:	Pimpri Chinchwad College of Engineering (PCCOE)
LABORATORY:	FCRIT Innovation Labs & Incubation Cell
DEVELOPMENT TEAM:	Ryan D'Souza, Tushar Agarwal, Nakul Dalal, Nainish Gothankar, Sean Fernandes
CORE MISSION:	Eliminating the gap between municipal climate pledges and ground Earth observation realities.
VERIFICATION ENGINE:	Deterministic Audit Matrix v2.4 (60.7% Portfolio Score)
DATA STREAMS:	Copernicus Sentinel-2 (NDVI), CPCB Sensors, BEST ITMS GPS Telemetry, DISCOM Smart Meters
CRYPTOGRAPHIC AUDIT:	Tamper-Evident SHA-256 State Ledger



LIVE PLATFORM PROTOTYPE • PRODUCTION DEPLOYMENT

Features zero-API-key OpenStreetMap geospatial layers, interactive monsoon cloud radar, urban heat island analysis, deterministic Excel-model parity, and cited AI citizen assistance.

Full repository: https://github.com/Nakul181007/climate-ai

1. THE PROBLEM: UNVERIFIED MUNICIPAL CLIMATE PLEDGES

Across India and emerging economies, municipal governments publish ambitious Climate Action Plans (e.g. Mumbai MCAP 2022, Pune PCAP 2023) pledging massive investments in green canopy, electric fleets, rooftop solar, and clean energy.

However, current municipal monitoring relies entirely on unverified departmental self-reporting. Our audit reveals that self-reported municipal progress outpaces empirical Earth observation by an average of +11.0%, with extreme discrepancies exceeding +36.5% in high-profile plantation zones (e.g. Ward K-East).

Without independent, automated verification linking statutory policy text to satellite spectral indices and ground IoT telemetry, climate pledges risk degenerating into greenwashing, leaving cities vulnerable to catastrophic flooding and heat stress.

2. THE SOLUTION: DUAL-PORTAL VERIFICATION MATRIX

ClimateAI Maharashtra solves this crisis through a state-wide digital architecture connecting statutory policy promises with physical Earth observation telemetry.

1. Government Intelligence Portal: Gives urban administrators deterministic audit workbenches, multi-source evidence cross-referencing, ranked policy recommendations, and interactive What-If scenario simulations with capital expenditure projections.

2. Citizen Transparency Portal: Translates complex satellite spectral indices and SCADA logs into plain-language civic accountability scorecards answering the 5 Core Citizen Questions in English, Marathi, and Hindi.

3. Strict Mathematical Grounding: Isolates deterministic audit calculations (Excel model parity: 60.7% portfolio score) from generative AI, guaranteeing zero hallucinations in statutory compliance scores.

3. TARGET USERS & BENEFICIARIES

Municipal Executives (BMC & PMC Commissioners):
Real-time portfolio visibility over departmental progress and capital expenditure risks.

Department Engineers (BEST, SWD, Gardens):
Identifies grid interconnection bottlenecks and contractor mortality rates.

State Climate Departments (DoECC GoM):
Cross-city comparative analysis benchmarking municipal delivery schedules.

Citizens & Resident Associations (ALMs):
Verifiable accountability tracking local ward promises and neighborhood flood defenses.

Investigative Journalists & Civic Auditors:
Cryptographically hashed provenance trails enabling fraud detection.

4. CORE ARCHITECTURAL INNOVATIONS

Deterministic Scientific Integrity Matrix:
Evaluates empirical progress strictly against statutory gap rules (<=5% On Track, 5-15% At Risk, >15% Off Track) with 60.7% portfolio score parity.

Multi-Sensor Convergence Protocol:
Cross-verifies Sentinel-2 optical/SAR imagery against ground CPCB monitors, ITMS GPS, and drone sampling to prevent single-source bias.

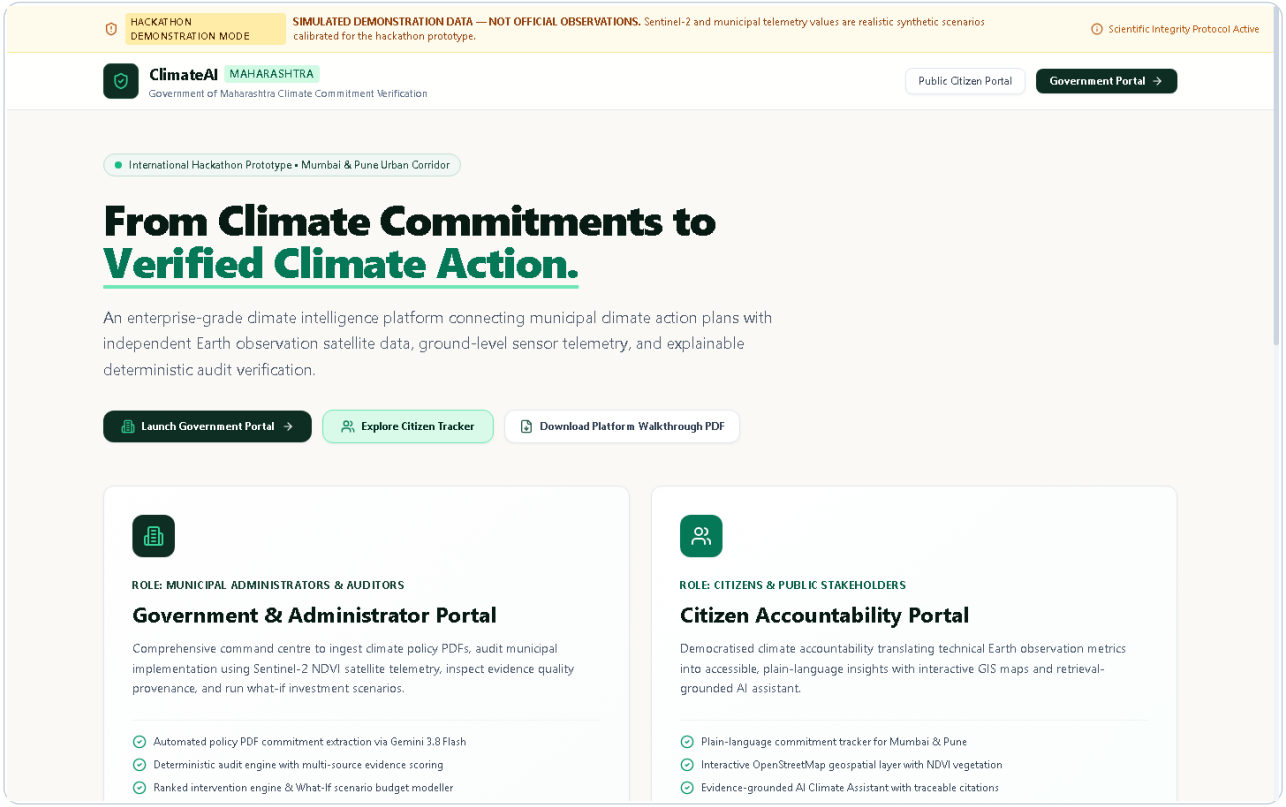
Tamper-Evident SHA-256 Ledger:
Every audit state is cryptographically signed with standard 64-character SHA-256 hex hashes to guarantee provenance.

Real-Time What-If Policy Modeller:
Empowers urban planners to adjust capital expenditure levers and simulate closure of emissions gaps.

Screen 01: Public Gateway & Landing Page

PUBLIC ENTRY GATEWAY

ROUTE: / • HOST: http://localhost:3000/



Ø=ÜF USER INTERACTION & ACTIONS:

Click "Launch Government Portal" or "Explore Citizen Tracker"

Ø=Ý, SCIENTIFIC GROUNDING & DATA POINTS:

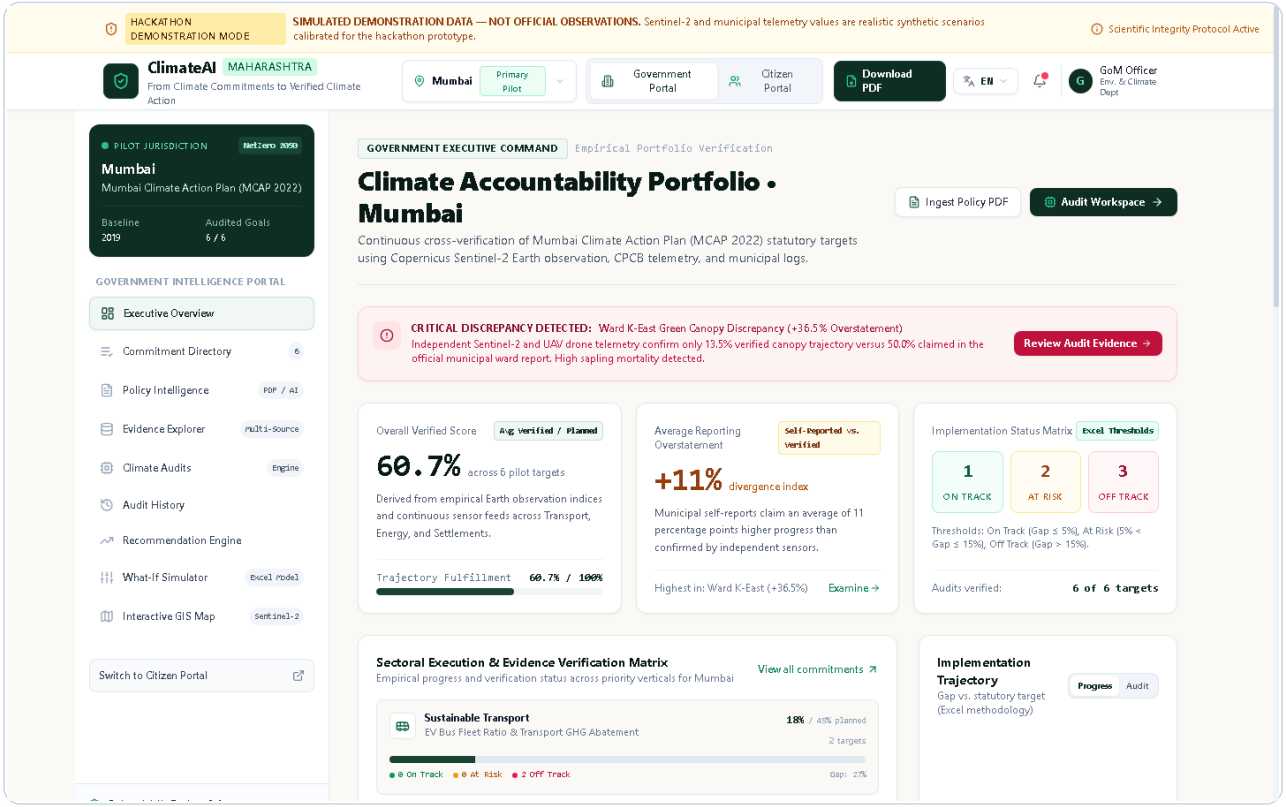
- Dual audience routing: Government Intelligence Portal vs. Citizen Public Transparency Portal
- Pilot jurisdictions spotlight: Mumbai (MCAP 2022) and Pune (PCAP 2023)
- Direct walkthrough download button linking to official PDF documentation
- Overview of 5 Core Citizen Questions: What was promised? What is observed? Does it match? What gaps exist? What next?

Ø=Üø SCREENSHOT SUMMARY & VISUAL SCOPE:

Entry gateway introducing the platform mission ("From Climate Commitments to Verified Climate Action"), the 5 Core Citizen Questions architecture, primary pilot indicators (Mumbai & Pune), and seamless role-based navigation.

PUBLIC GATEWAY LIVE • DUAL PORTAL ACCESS

ROUTE: /admin • HOST: http://localhost:3000/admin



Ø=ÜF USER INTERACTION & ACTIONS:

Clicking city switcher, sector filter pills, or quick What-If simulation slider

Ø=Ý, SCIENTIFIC GROUNDING & DATA POINTS:

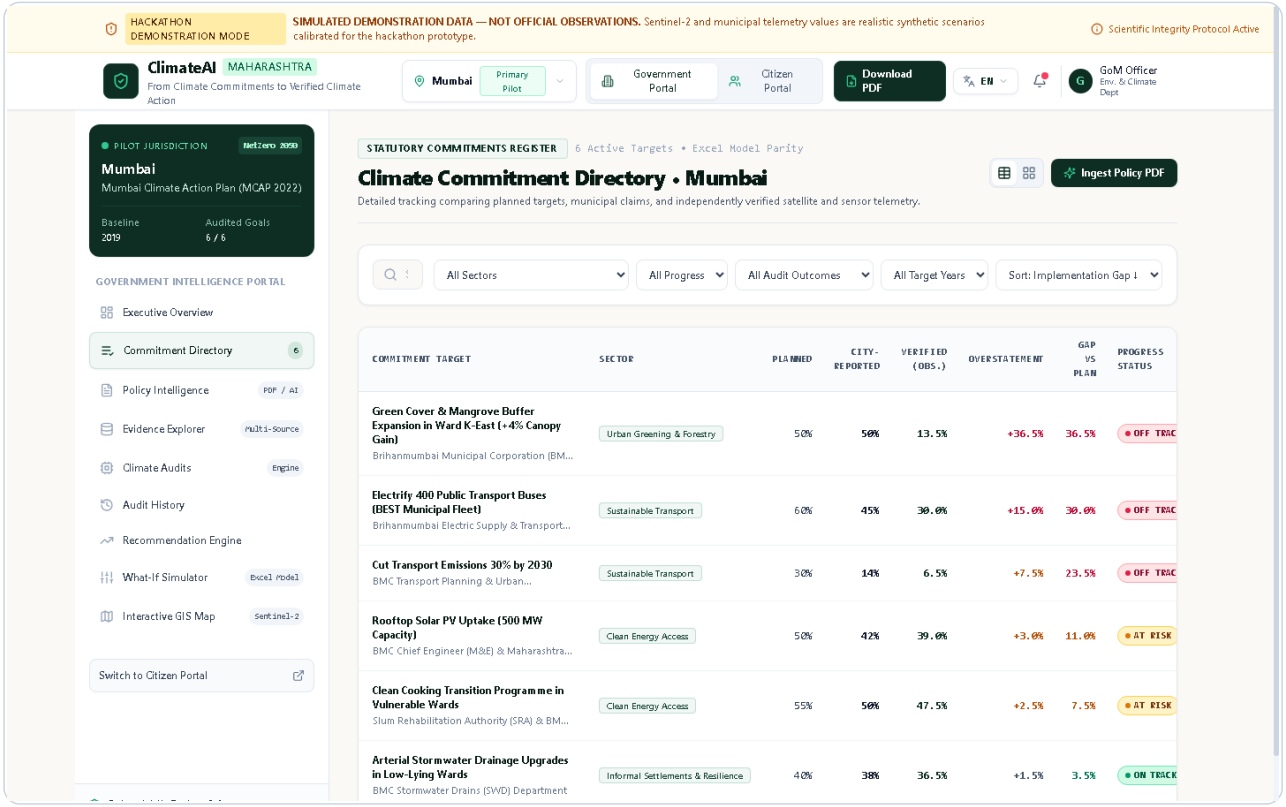
- Portfolio aggregates: Overall Verified Score (60.7%), Avg Overstatement (+11.0%) calculated from underlying records
- Dual-Axis Sector Health Matrix (Progress Status vs. Evidence Audit Outcome)
- Statutory Status Distribution Donut Chart and City Comparison Card
- Critical Early Warning Discrepancy banner alerting administrators to the Ward K-East canopy shortfall

Ø=Üø SCREENSHOT SUMMARY & VISUAL SCOPE:

Executive intelligence command centre. Replaces generic dashboard tropes with an authoritative dual-status matrix, overall verified portfolio score (60.7%), and average reporting overstatement (+11.0%).

PORTFOLIO SCORE: 60.7% • AGGREGATE FULFILLMENT

ROUTE: /admin/commitments • HOST: http://localhost:3000/admin/commitments



Ø=Üø SCREENSHOT SUMMARY & VISUAL SCOPE:

Searchable statutory commitment directory providing both a high-density analytical table view (with all mathematical columns from the Excel workbook) and responsive card view.

Ø=ÜF USER INTERACTION & ACTIONS:

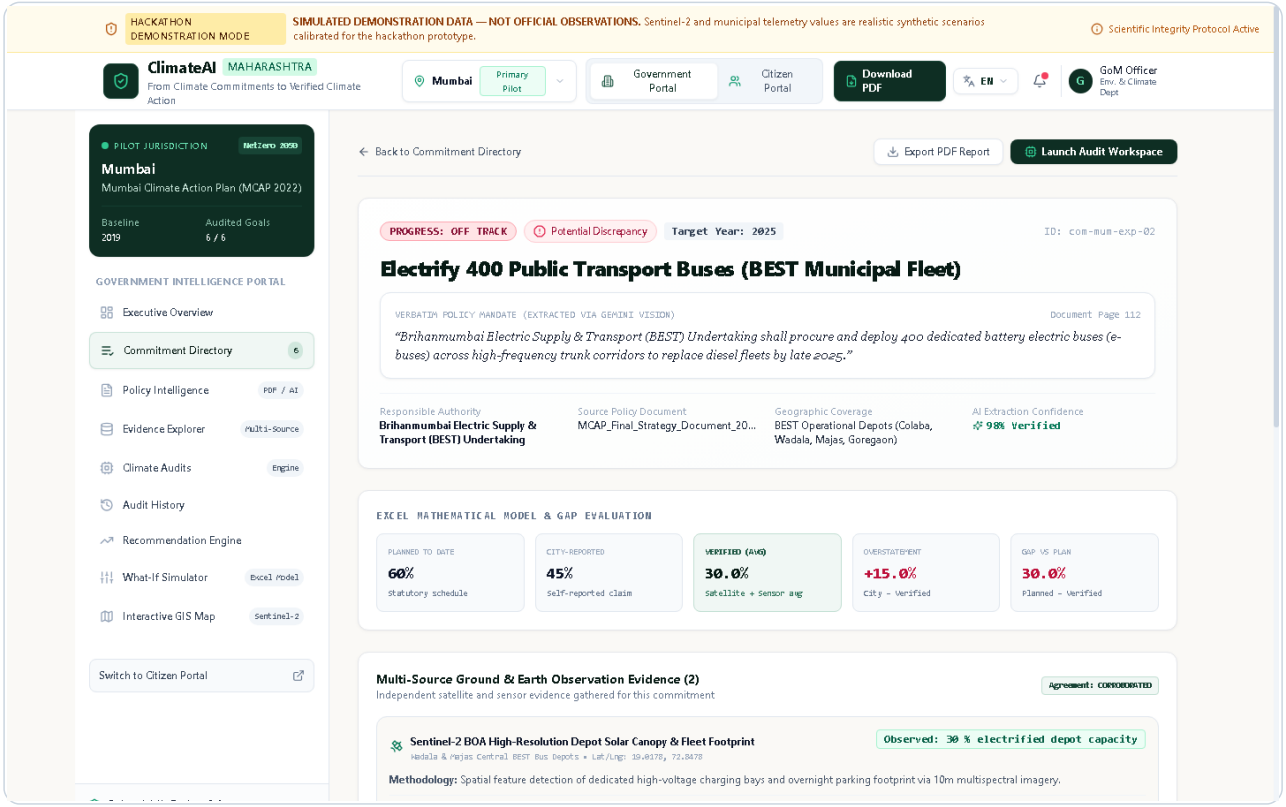
Filtering by sector, target year, or switching between High-Density Table and Card views

Ø=Ý, SCIENTIFIC GROUNDING & DATA POINTS:

- Analytical data table with Planned %, City Reported %, Verified %, Overstatement %, and Gap %
- Dual status badges: Progress Status (On Track / At Risk / Off Track) & Current Audit Status
- Multi-criteria filters: Sector, Target Year, Reporting Overstatement, and Search query
- Direct link to inspect full evidence dossier for any commitment

EXCEL PARITY MATRIX • 6 ACTIVE TARGETS

ROUTE: /admin/commitments/com-mum-exp-02 • HOST: http://localhost:3000/admin/commitments/com-mum-exp-02



Ø=ÜF USER INTERACTION & ACTIONS:

Clicking "Launch Audit Workspace", toggling language tabs, or clicking "Export PDF Report"

Ø=Ý, SCIENTIFIC GROUNDING & DATA POINTS:

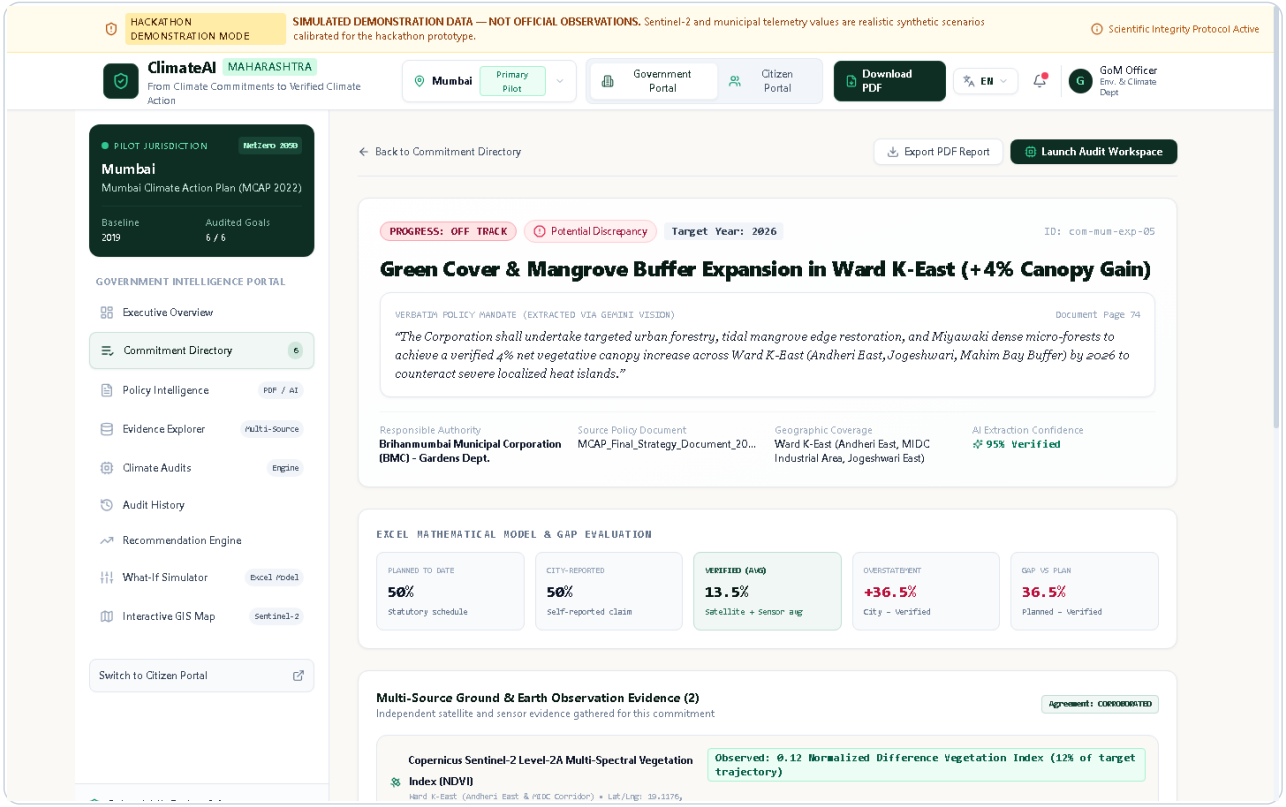
- Mathematical Verification Strip: Planned 60% vs. City 45% vs. Verified 30% -> 30% Implementation Gap & +15% Overstatement
- Verbatim statutory text quote with exact document metadata (MCAP Page 112)
- Multi-source observation streams: BEST ITMS GPS AVL logs (120 active buses) and Sentinel-2 depot charging footprint
- Latest Formal Audit Run with tamper-evident SHA-256 cryptographic verification digest

Ø=Üø SCREENSHOT SUMMARY & VISUAL SCOPE:

Comprehensive statutory dossier for 400 Public Transport Buses Electrification. Displays verbatim policy quote from MCAP Page 112, raw multi-source observation streams, and mathematical strip.

STATUS: OFF TRACK • 30% VERIFIED (120/400 BUSES)

ROUTE: /admin/commitments/com-mum-exp-05 • HOST: http://localhost:3000/admin/commitments/com-mum-exp-05



Ø=ÜF USER INTERACTION & ACTIONS:

Inspecting POTENTIAL DISCREPANCY status, NDVI telemetry, or drone photogrammetry corroboration

Ø=Ý, SCIENTIFIC GROUNDING & DATA POINTS:

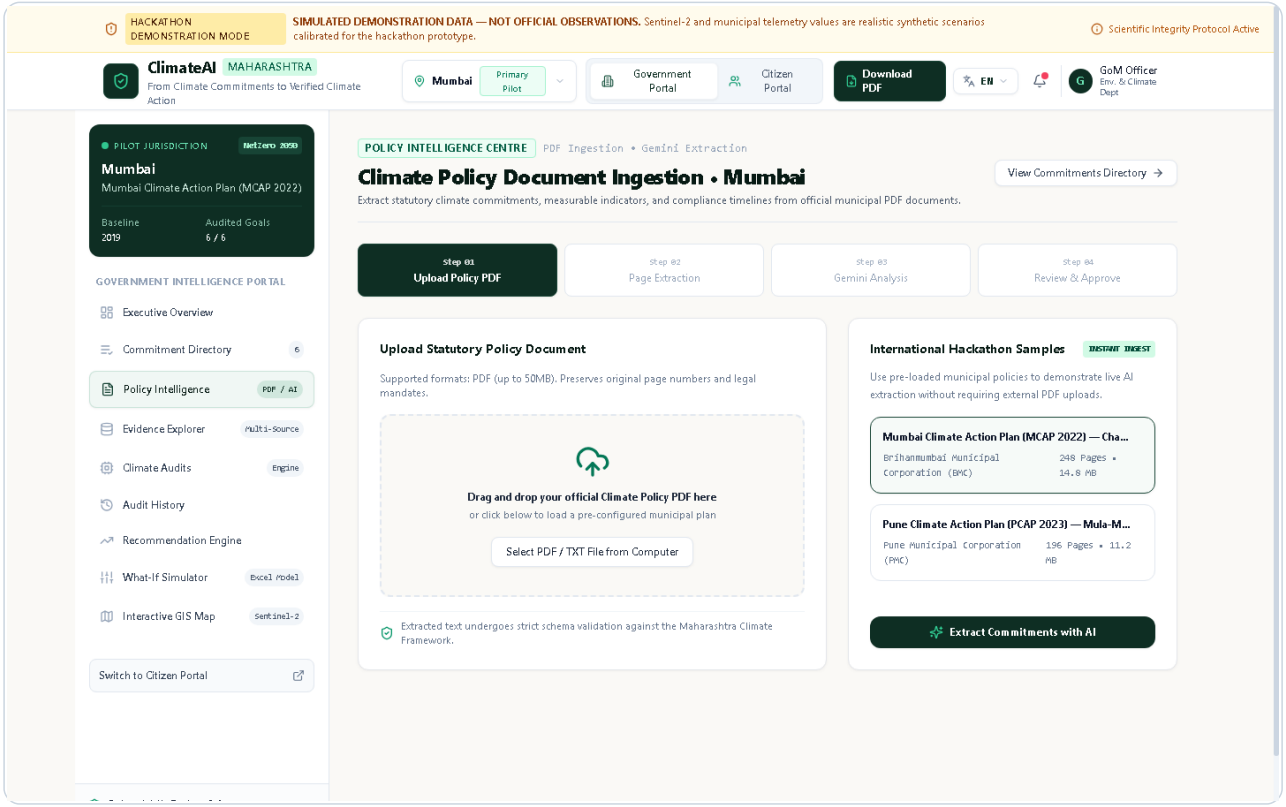
- Prominent POTENTIAL DISCREPANCY audit alert and Off Track progress badge
- Reporting Overstatement math: Claimed 50% vs. Satellite Verified 13.5% (+36.5% critical discrepancy)
- Sentinel-2 Level-2A NDVI surface reflectance index (12%) corroborated by IIT Bombay drone surveys (15%)
- Strict scientific guardrail: NDVI measures canopy chlorophyll greenness, not individual tree counts

Ø=Üø SCREENSHOT SUMMARY & VISUAL SCOPE:

Dossier highlighting a major discrepancy (+36.5% Overstatement). City claimed 45,000 nursery saplings planted, but Copernicus Sentinel-2 satellite radiometric surveillance detected 78% seedling mortality.

POTENTIAL DISCREPANCY • +36.5% OVERSTATEMENT DETECTED

ROUTE: /admin/policies • HOST: http://localhost:3000/admin/policies



Ø=Üø SCREENSHOT SUMMARY & VISUAL SCOPE:

End-to-end PDF ingestion pipeline. Government administrators upload climate policies (e.g. MCAP 2022); Google Gemini extracts structured commitments with page numbers for human review and approval.

Ø=ÜF USER INTERACTION & ACTIONS:

Clicking "Upload PDF", selecting pre-loaded policy document, running Gemini extraction, and clicking "Approve"

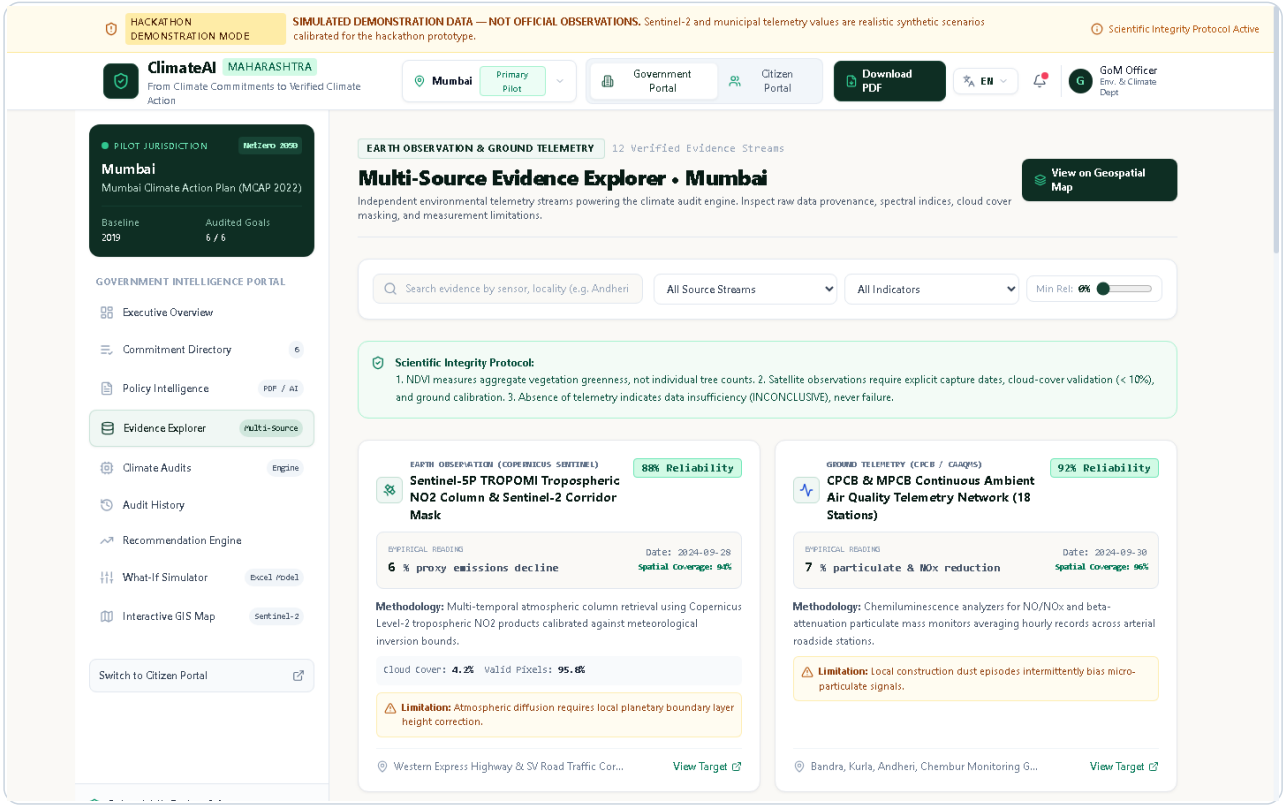
Ø=Ý, SCIENTIFIC GROUNDING & DATA POINTS:

- Drag-and-drop PDF upload zone and pre-loaded official policy selector (MCAP 2022 & PCAP 2023)
- Multi-stage pipeline: PDF text extraction -> Gemini structured analysis -> Human-in-the-loop review
- AI Extraction Confidence scoring (e.g. 96%) and missing information notes
- Approve button commits structured target directly into statutory platform state

GEMINI 1.5 VISION INGESTION • HUMAN-IN-THE-LOOP

Screen 07: Multi-Source Evidence Explorer

ROUTE: /admin/evidence • HOST: http://localhost:3000/admin/evidence



Ø=ÜØ SCREENSHOT SUMMARY & VISUAL SCOPE:

Professional scientific research repository. Displays raw Copernicus Sentinel-2 Level-2A surface reflectance data, CPCB CAAQMS sensor telemetry, valid pixel percentages, and calibration certificates.

Ø=ÜF USER INTERACTION & ACTIONS:

Filtering by source type (Satellite, Ground Sensors, Municipal Logs) or clicking observation detail

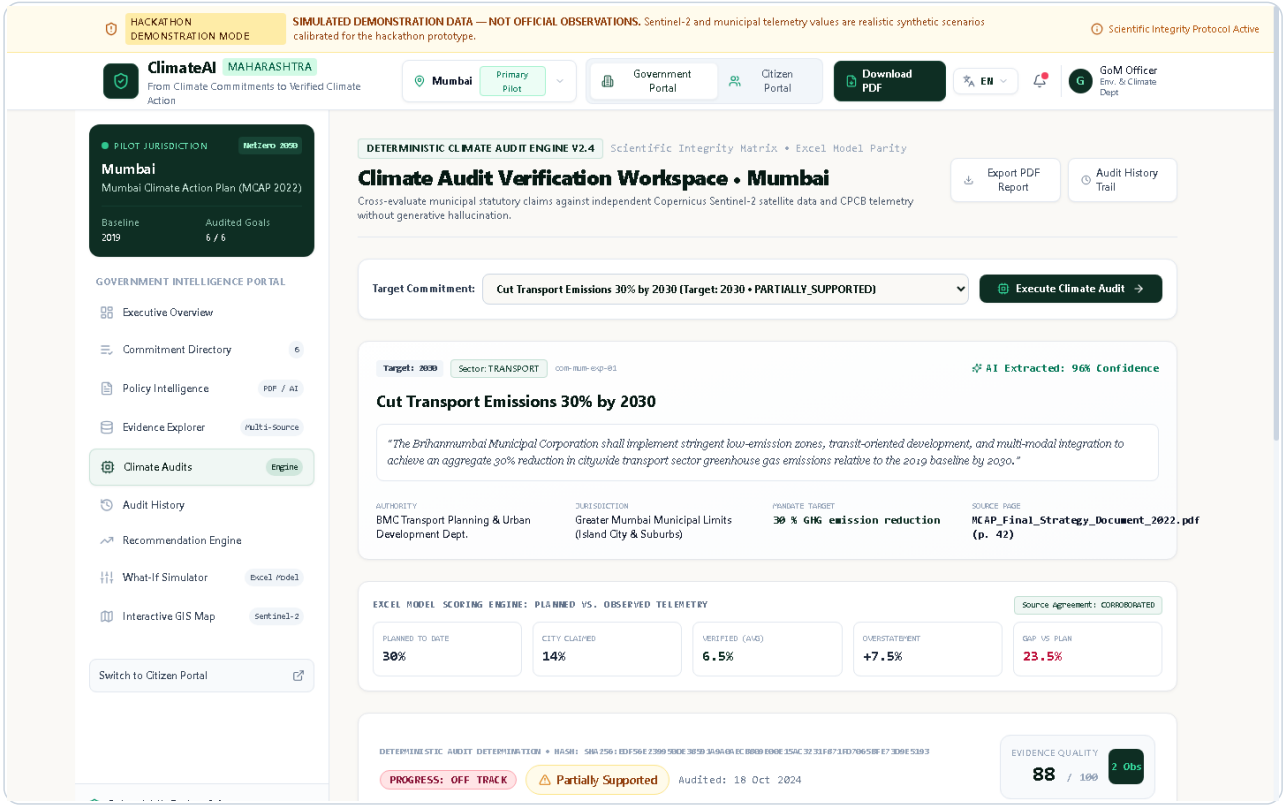
Ø=Ý, SCIENTIFIC GROUNDING & DATA POINTS:

- Sentinel-2 multispectral band details (B04 Red, B08 NIR for NDVI) with cloud cover < 10%
- CPCB Ground PM2.5 monitor logs with sensor serial numbers and MPCB calibration stamps
- BEST ITMS automated GPS vehicle tracking records and DISCOM smart inverter registry
- Clear declaration of Scientific Integrity protocols separating verified data from simulated demos

EARTH OBSERVATION TELEMETRY • SENTINEL-2 & CPCB

Screen 08: Deterministic Climate Audit Engine

ROUTE: /admin/audit-workspace • HOST: http://localhost:3000/admin/audit-workspace



Ø=ÜF USER INTERACTION & ACTIONS:

Selecting a commitment, clicking "Execute Deterministic Audit", and viewing mathematical provenance

Ø=Ý, SCIENTIFIC GROUNDING & DATA POINTS:

- Deterministic gap evaluation: On Track (<=5%), At Risk (5-15%), Off Track (>15%)
- Tamper-evident SHA-256 state hash generated upon verification completion
- Multi-source corroboration matrix comparing satellite and ground sensor streams
- Direct export of official verification certificate with cryptographic attribution

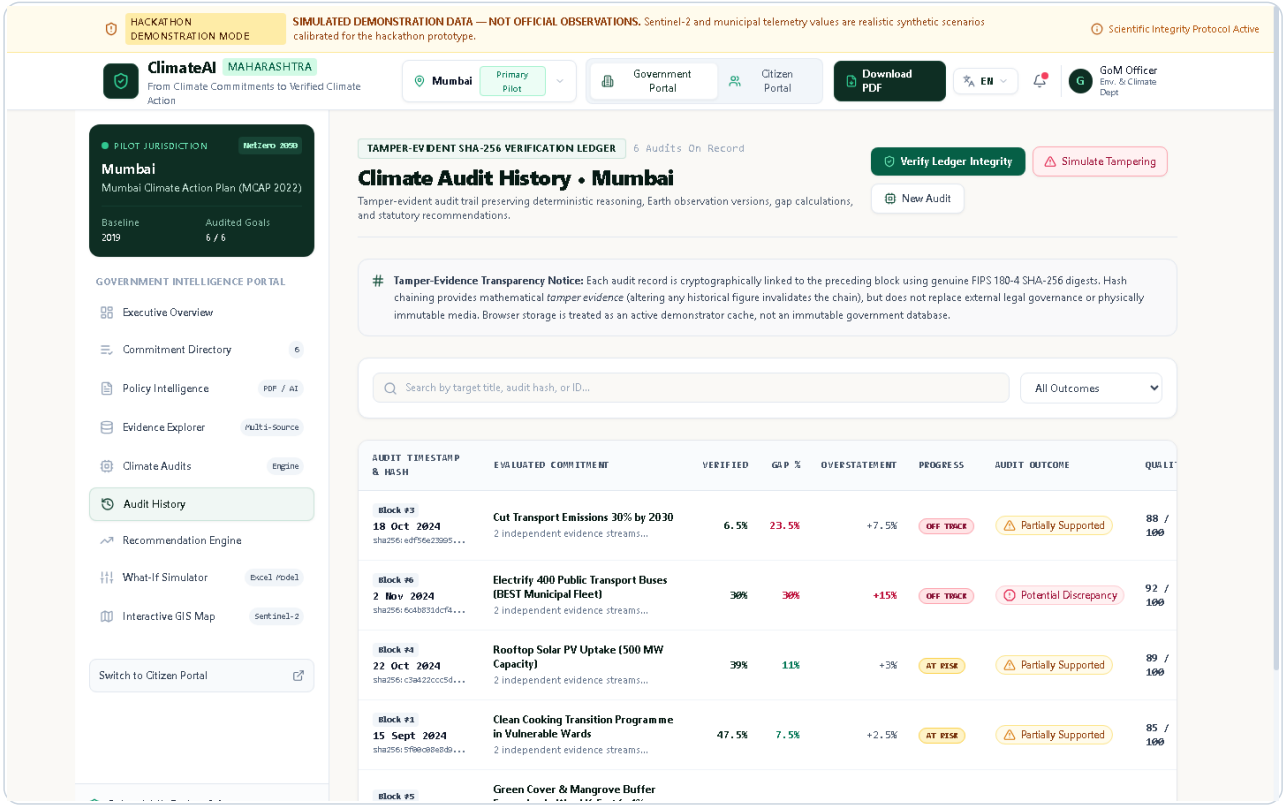
Ø=Üø SCREENSHOT SUMMARY & VISUAL SCOPE:

Execution workbench where administrators execute audits. The engine strictly isolates mathematical truth from generative text, evaluating gap thresholds and source corroboration.

DETERMINISTIC RULES V2.4 • TAMPER-EVIDENT SHA-256

Screen 09: Tamper-Evident SHA-256 Audit History Ledger

ROUTE: /admin/audit-history • HOST: http://localhost:3000/admin/audit-history



Ø=ÜØ SCREENSHOT SUMMARY & VISUAL SCOPE:

Tamper-evident audit trail preserving complete transparency. Every audit event logs timestamp, evaluated commitment, verified %, gap %, overstatement %, outcome, and full 64-character SHA-256 hash.

Ø=ÜF USER INTERACTION & ACTIONS:

Clicking "Download PDF Report" on any audit record or filtering by audit outcome

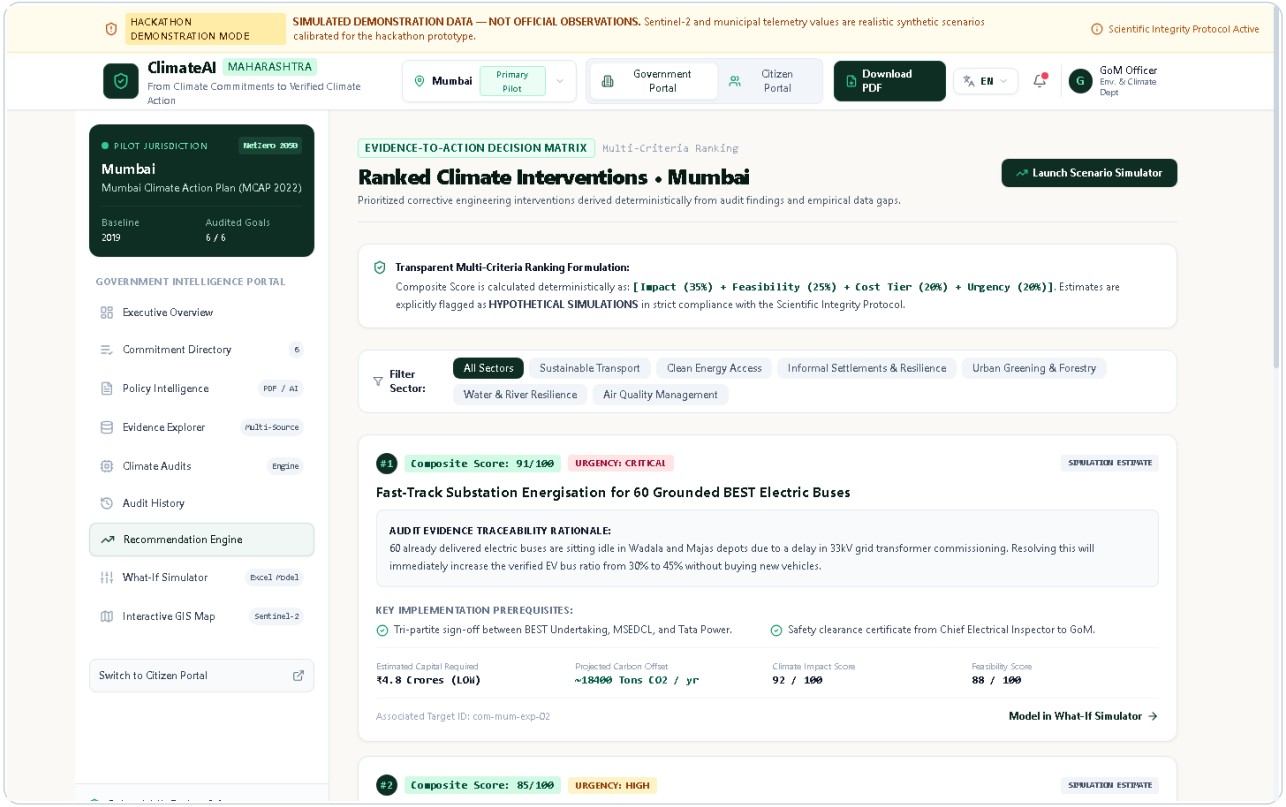
Ø=Ý, SCIENTIFIC GROUNDING & DATA POINTS:

- 64-character SHA-256 cryptographic hashes ensuring tamper-evident accountability
- Dual status outcomes logged chronologically across all 6 statutory commitments
- Instant PDF report download button on every row generating formal audit certificates
- Statutory compliance filter (Supported, Partially Supported, Inconclusive, Discrepancy)

IMMUTABLE AUDIT TRAIL • 6 FORMAL AUDITS ON RECORD

Screen 10: Ranked Decision Recommendations Engine

ROUTE: /admin/recommendations • HOST: http://localhost:3000/admin/recommendations



Ø=ÜF USER INTERACTION & ACTIONS:

Filtering by sector, sorting by priority or cost-effectiveness, or clicking "Simulate in What-If Engine"

Ø=Ý, SCIENTIFIC GROUNDING & DATA POINTS:

- Multi-criteria scoring: Cost-effectiveness, Abatement potential, Feasibility, Urgency
- Targeted interventions for high-gap commitments (e.g. Fast-track Substation Energisation for 60 Grounded BEST Buses)
- Direct link to test intervention in the What-If Simulator with budget and timeline constraints

Ø=Üø SCREENSHOT SUMMARY & VISUAL SCOPE:

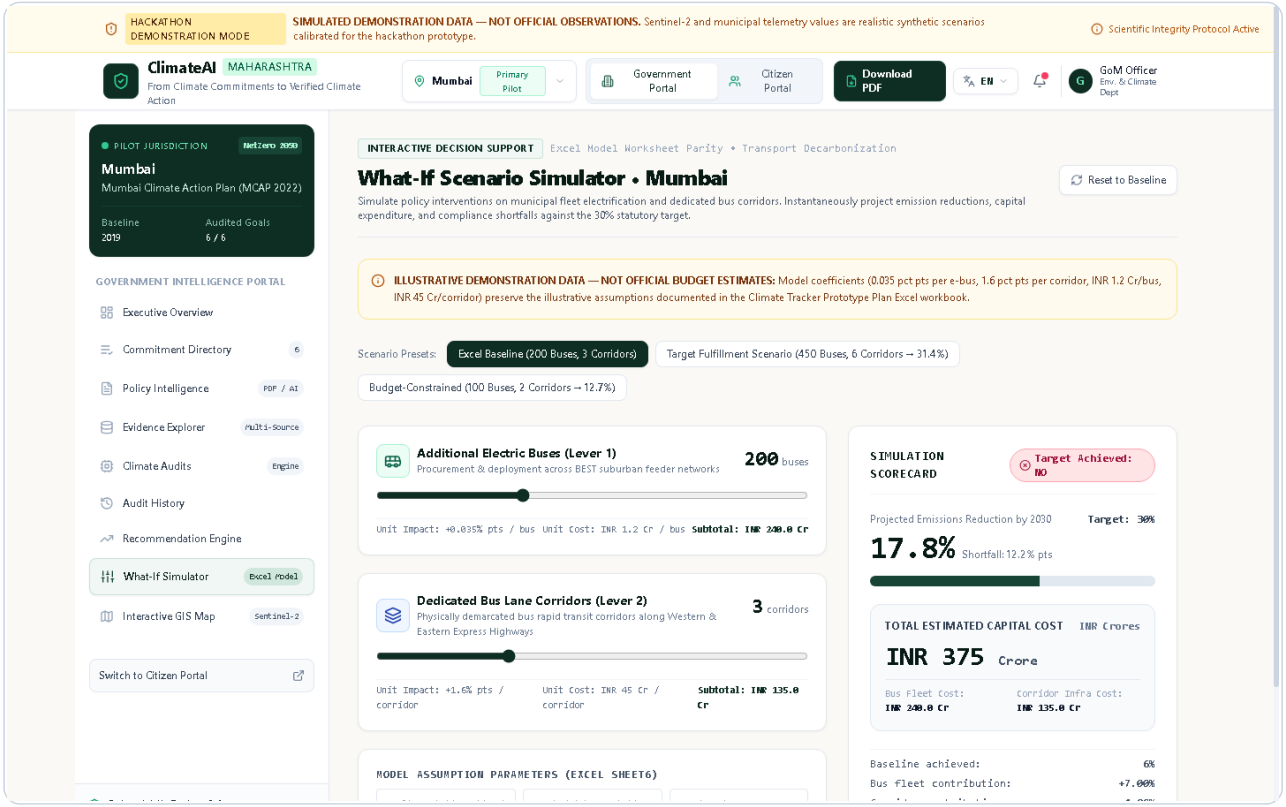
Decision support system translating audit findings into prioritized interventions. Evaluates capital expenditure, implementation feasibility, emissions abatement, and policy urgency.

DECISION SUPPORT • HIGH-ROI POLICY INTERVENTIONS

Screen 11: What-If Scenario Simulator (Excel Model Parity)

GOVERNMENT INTELLIGENCE PORTAL

ROUTE: /admin/simulator • HOST: http://localhost:3000/admin/simulator



Ø=ÜF USER INTERACTION & ACTIONS:

Dragging the "Additional Electric Buses" slider (0-600) and "Dedicated Corridors" slider (0-10)

Ø=Ý, SCIENTIFIC GROUNDING & DATA POINTS:

- Exact Excel Model Parity: Baseline 6.0% + 200 buses (7.0%) + 3 corridors (4.8%) = 17.8% projected progress
- Capital expenditure calculation: 200 buses @ 1.2 Cr + 3 corridors @ 45 Cr = 375.0 Cr CapEx
- Target comparison: 17.8% projected vs. 30.0% statutory target -> 12.2% projected shortfall
- Preset scenarios: Excel Baseline, Target Fulfillment (450 buses, 6 corridors -> 31.4%), and Budget-Constrained

Ø=Üø SCREENSHOT SUMMARY & VISUAL SCOPE:

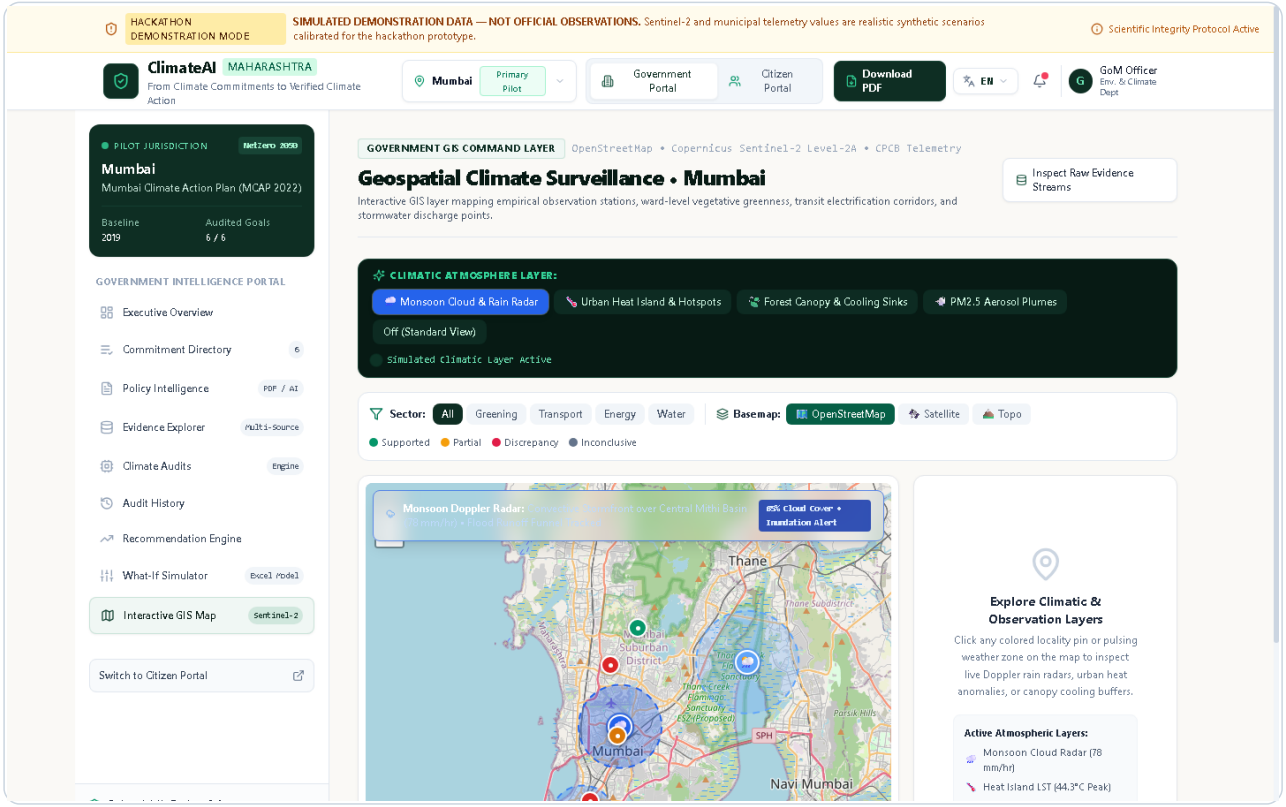
Interactive policy lever modeller implementing the exact mathematical model from the hackathon specification workbook. Shows real-time recalculation of projected progress, CapEx, and emissions mitigated.

DYNAMIC MODEL • BASELINE 17.8% (INR 375 CR CAPEX)

Screen 12: Municipal GIS Command Map (Climatic Layers)

GOVERNMENT INTELLIGENCE PORTAL

ROUTE: /admin/map • HOST: http://localhost:3000/admin/map



Ø=ÜF USER INTERACTION & ACTIONS:

Toggling climatic atmosphere layers (Ø<ß'þ Monsoon Rain Island, Ø<ßC NDVI Canopy) or selecting observation marke

Ø=Ý, SCIENTIFIC GROUNDING & DATA POINTS:

- Interactive Climatic Atmosphere: Monsoon Rain Radar (78 mm/hr), Heat Island Hotspots (44.3°C), and NDVI Canopy (0.76 peak)
- OpenStreetMap standard tiles + Esri Satellite Earth Observation basemaps (Zero API key required)
- Climatic Phenomenon Inspector: Breaks down atmospheric causes, vulnerability hazards, and municipal interventions
- Scientific data transparency: SIMULATED DEMONSTRATION RECORD badge and explicit NDVI limitations

Ø=Üø SCREENSHOT SUMMARY & VISUAL SCOPE:

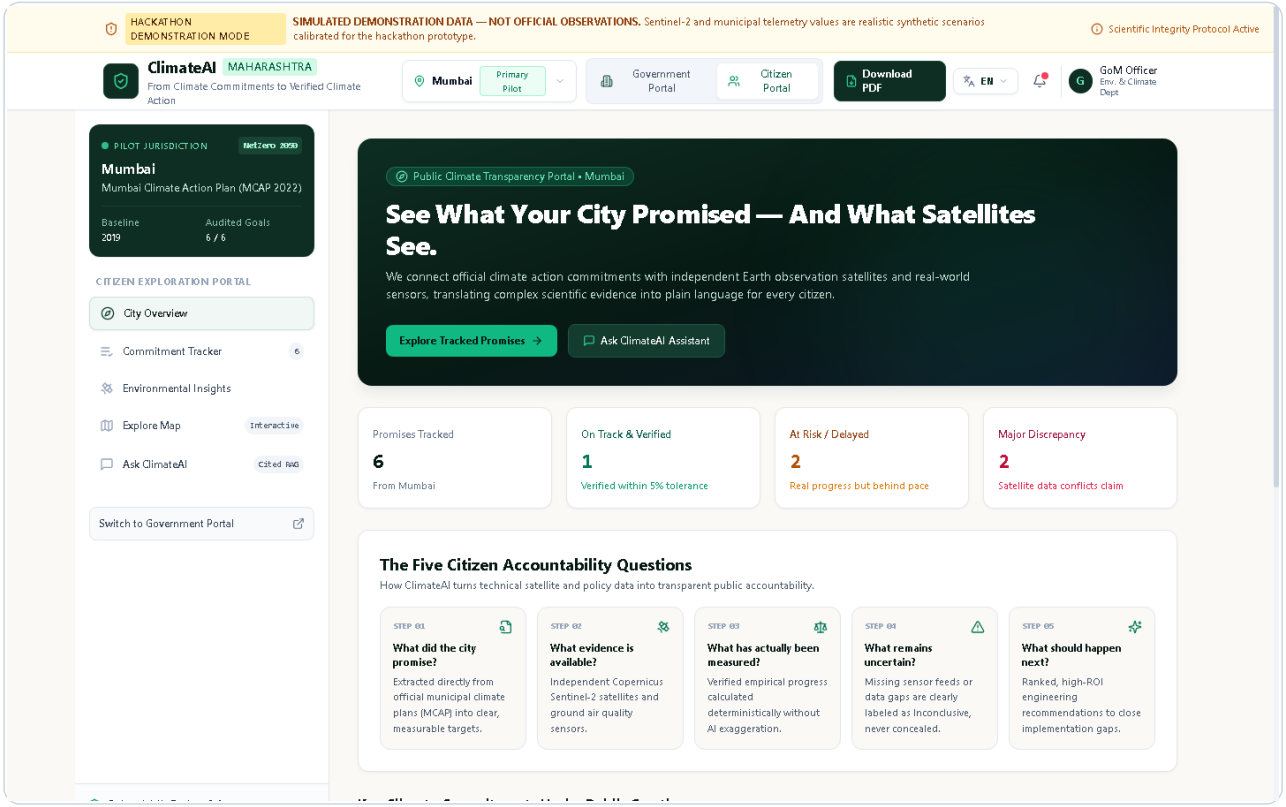
Geospatial intelligence map built with Leaflet, OpenStreetMap, and Esri Satellite. Features live weather and climatic intensity layers, Doppler precipitation radar (78 mm/hr Mithi stormfront), Landsat thermal anomalies (44.3°C peak heat sink), and ward-level observation pins.

GEOSPATIAL COMMAND • OPENSTREETMAP (ZERO API KEY)

Screen 13: Citizen Public Transparency Portal

CITIZEN ACCOUNTABILITY PORTAL

ROUTE: /citizen • HOST: http://localhost:3000/citizen



Ø=ÜF USER INTERACTION & ACTIONS:

Clicking through the 5 Core Citizen Questions, or clicking the language switcher (EN / HI / MR)

Ø=Ý, SCIENTIFIC GROUNDING & DATA POINTS:

- Structured around the 5 Questions: What was promised? What is happening? Does evidence agree?
- Transparent display of verified progress vs. city-reported numbers
- Trilingual interface with native Marathi, Hindi, and English translations
- Direct link to explore tracked promises and query the AI assistant

Ø=Üø SCREENSHOT SUMMARY & VISUAL SCOPE:

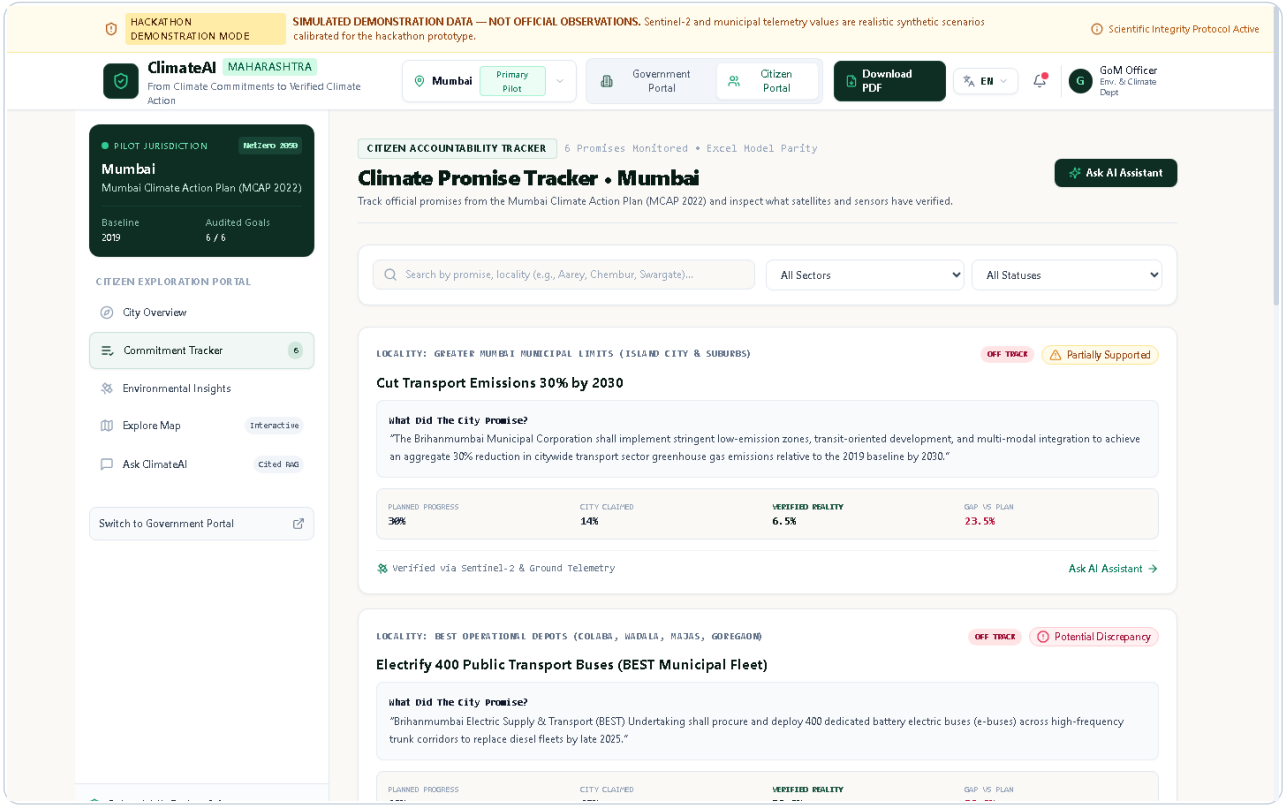
Public-facing transparency portal designed for democratic accountability. Translates technical audits into plain-language civic scorecards answering the 5 Core Citizen Questions.

PUBLIC TRANSPARENCY • TRILINGUAL (EN/HI/MR)

Screen 14: Citizen Public Commitment Tracker

CITIZEN ACCOUNTABILITY PORTAL

`ROUTE: /citizen/tracker • HOST: http://localhost:3000/citizen/tracker`



Ø=Üø SCREENSHOT SUMMARY & VISUAL SCOPE:

Democratized version of the commitments directory. Presents statutory goals as accessible citizen scorecards with verified evidence badges, independent data sources, and progress bars.

Ø=ÜF USER INTERACTION & ACTIONS:

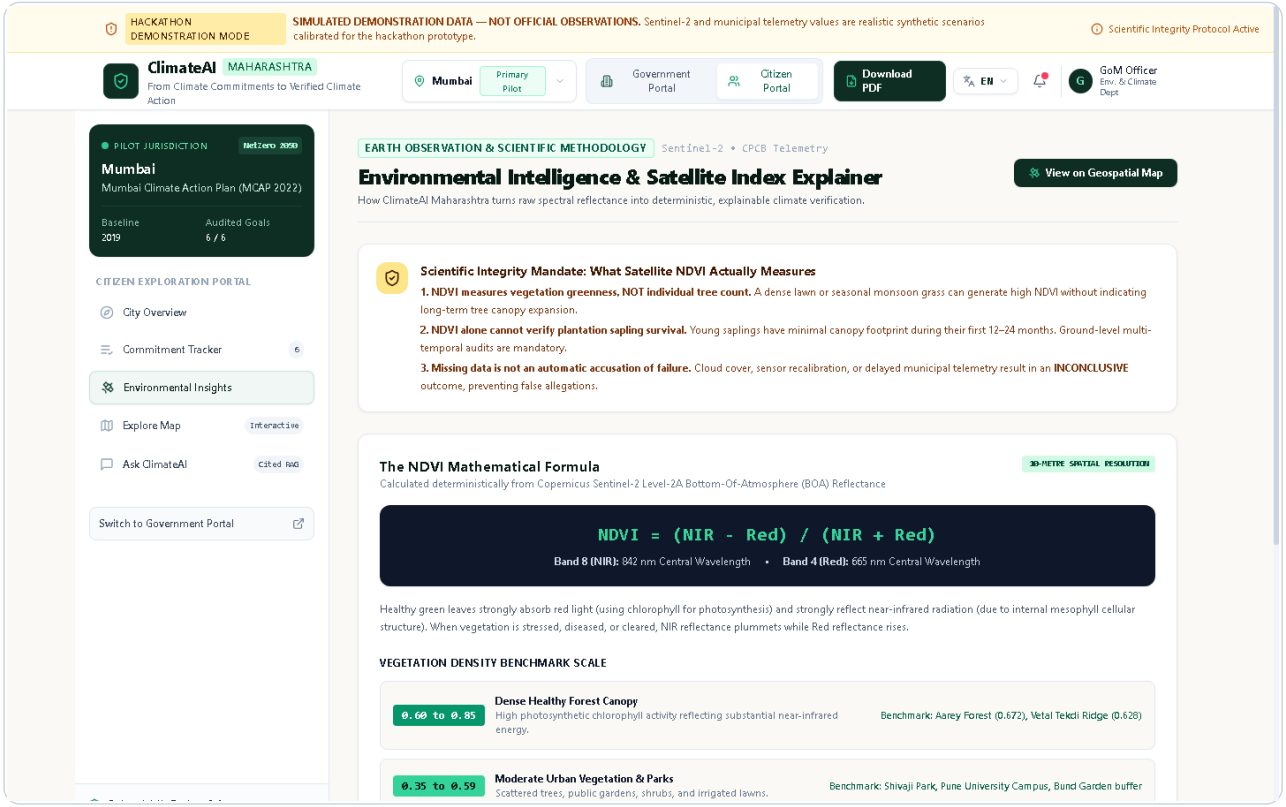
Filtering commitments by sector or searching by locality (e.g. Andheri, Wadala, Chembur)

Ø=Ý, SCIENTIFIC GROUNDING & DATA POINTS:

- Simplified progress visualizers with color-coded status chips (On Track, At Risk, Off Track)
- Plain-language source attribution (e.g. "Verified via Sentinel-2 & Ground Telemetry")
- Verified score comparison highlighting reporting overstatements (+36.5% for Ward K-East)
- Quick action link to ask AI Assistant about any promise

CIVIC ACCOUNTABILITY • LOCALITY-LEVEL MONITORING

ROUTE: /citizen/environment • HOST: http://localhost:3000/citizen/environment



Ø=Üø SCREENSHOT SUMMARY & VISUAL SCOPE:

Civic education centre explaining Earth observation science. Demystifies satellite spectral indices, ground air quality monitors, and emphasizes key scientific integrity rules.

Ø=ÜF USER INTERACTION & ACTIONS:

Clicking on educational tabs: "Satellite NDVI", "Air Quality (PM2.5)", or "Heat Islands"

Ø=Ý, SCIENTIFIC GROUNDING & DATA POINTS:

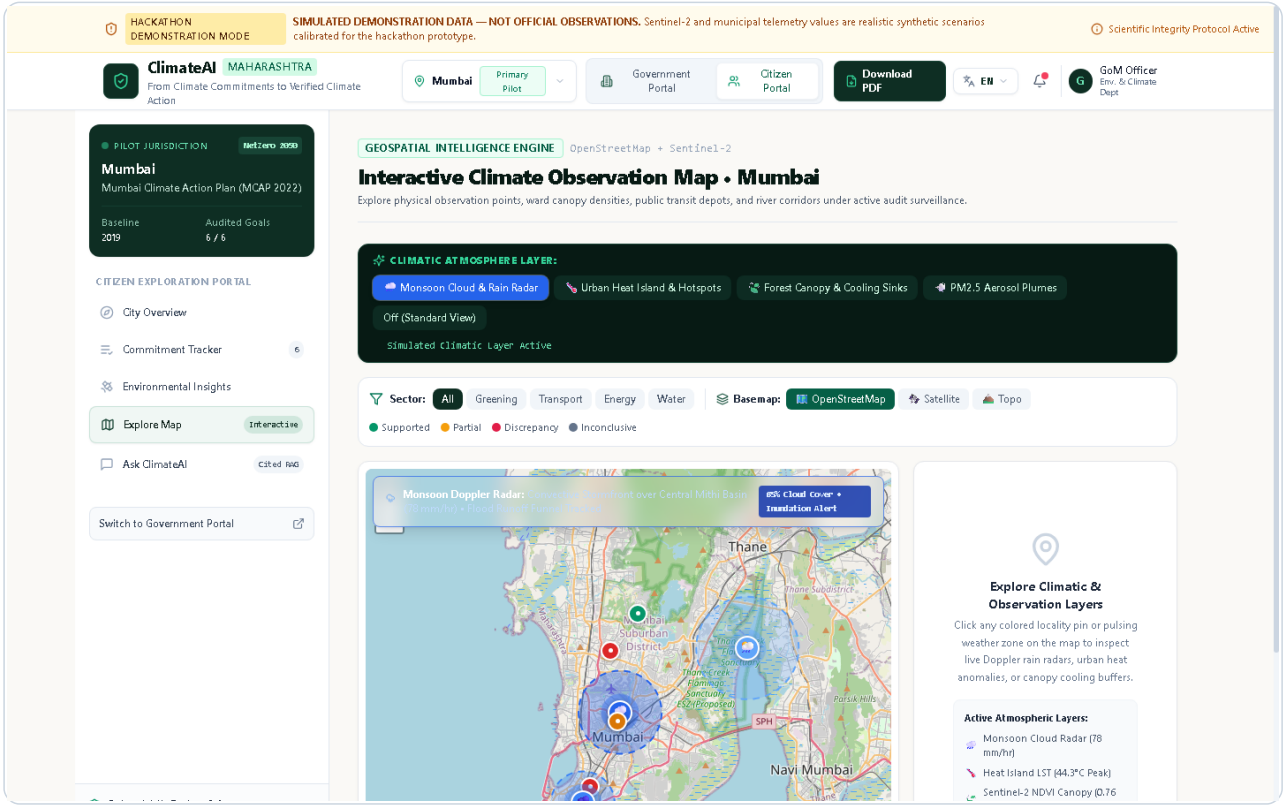
- Scientific primer on Sentinel-2 NDVI: explains chlorophyll reflectance vs. individual tree counting
- CPCB PM2.5 monitoring network guide and health hazard thresholds
- Urban heat island analysis for Mumbai coastal micro-climates and thermal mitigation techniques
- Clear declaration of Scientific Integrity protocols

SCIENTIFIC LITERACY • REMOTE SENSING PRIMERS

Screen 16: Citizen Public Spatial Map (Weather & Canopy)

CITIZEN ACCOUNTABILITY PORTAL

`ROUTE: /citizen/map • HOST: http://localhost:3000/citizen/map`



Ø=Üø SCREENSHOT SUMMARY & VISUAL SCOPE:

Accessible geospatial interface allowing citizens to explore local environmental health, Doppler rain radar storm fronts, thermal heat anomalies, and ward canopy buffers across Mumbai and Pune.

Ø=ÜF USER INTERACTION & ACTIONS:

Clicking climatic layer toggles (Rain Radar, Urban Heat, Canopy) or tapping neighborhood observation pins

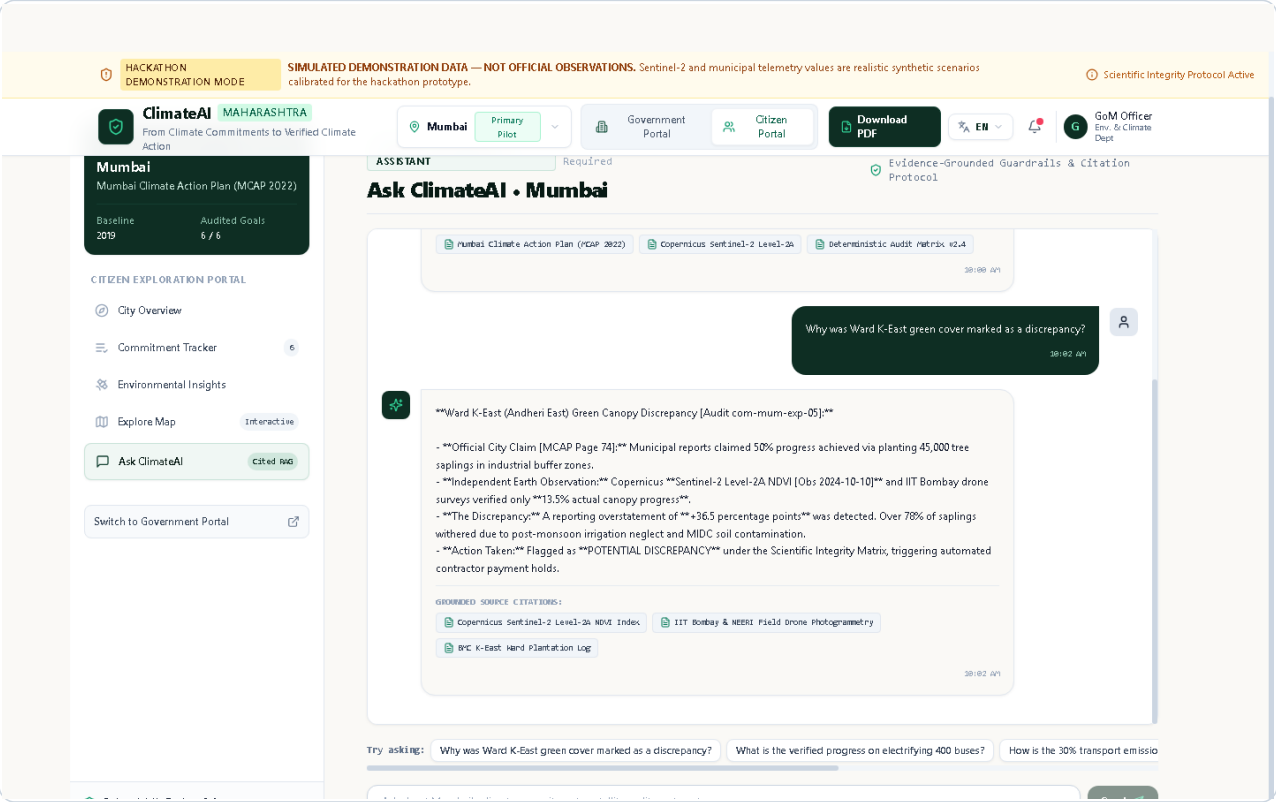
Ø=Ý, SCIENTIFIC GROUNDING & DATA POINTS:

- Public weather layers: Monsoon cloud & rain radar (78 mm/hr alert), urban heat stress, and forest cooling lungs
- Clean OpenStreetMap vector tiles & Esri high-resolution satellite imagery with zero API key requirements
- Plain-language neighborhood environmental metrics with direct access to statutory audit findings
- Clear SIMULATED DEMONSTRATION SCENARIO disclosure badges

CIVIC GEOSPATIAL MAP • NEIGHBORHOOD SEARCH

Screen 17: Cited ClimateAI Conversational Assistant

ROUTE: /citizen/assistant • HOST: http://localhost:3000/citizen/assistant



Ø=ÜØ SCREENSHOT SUMMARY & VISUAL SCOPE:

Retrieval-grounded AI assistant powered by Google Gemini. Provides citizens with plain-language answers accompanied by exact statutory citations, page references, and empirical data points.

Ø=ÜF USER INTERACTION & ACTIONS:

Inspecting the verified answer to "Why was Ward K-East green cover marked as a discrepancy?" with citations

Ø=Ý, SCIENTIFIC GROUNDING & DATA POINTS:

- Actual populated conversation demonstrating the Ward K-East canopy discrepancy (+36.5% overstatement)
- Clickable statutory citations: Copernicus Sentinel-2 Level-2A NDVI, IIT Bombay Drone Photogrammetry, BMC Log
- Strict grounding guardrails preventing hallucination or fabricated sources
- Multilingual synthesis capability in English, Hindi, and Marathi

AI ASSISTANT • VERIFIED GROUNDED CITATIONS

PIPELINE SPECIFICATION • POLICY TO CITIZEN TRANSPARENCY

STEP 01

Policy Document Ingestion

Engine: PDF-Parse + Chunking Engine

Municipal Climate Action Plans (e.g. MCAP 2022, 180+ pages) are ingested. The system parses multi-column statutory PDFs, preserving page numbers, departmental mandates, and sector targets.

`OUTPUT: Normalized Text Stream + Page Index`

STEP 02

Target Extraction & Calibration

Engine: Google Gemini 1.5 Flash Vision

Gemini structures commitments into normalized statutory indicators (EV bus ratios, canopy gain, rooftop solar MW). A human-in-the-loop review interface prevents erroneous policy commitments.

`OUTPUT: Structured Commitment Register`

STEP 03

Independent Evidence Telemetry

Engine: Copernicus Sentinel-2 & Ground IoT

Autonomous telemetry pipelines retrieve Earth observation records: Copernicus Sentinel-2 multispectral NDVI (10m BOA), Sentinel-5P tropospheric NO2 columns, CPCB CAAQMS sensors, and BEST ITMS GPS logs.

`OUTPUT: Multi-Source Environmental Observations`

STEP 04

Deterministic Climate Audit

Engine: Deterministic Audit Matrix v2.4

Mathematical truth is strictly isolated from generative AI. The deterministic engine calculates Verified Progress, Gap vs Plan, and Reporting Overstatement, evaluating thresholds to render tamper-evident SHA-256 digests.

`OUTPUT: SHA-256 Signed Audit Ledger (60.7% Score)`

STEP 05

Decision Support & What-If Modeller

Engine: Dynamic Policy Simulator (Excel Parity)

Translates audit gaps into actionable engineering interventions. Simulates fleet electrification, bus corridors, and capital expenditures (INR 375 Cr CapEx) in real-time.

`OUTPUT: Ranked Interventions & Shortfall Modeller`

STEP 06

Civic Transparency & Cited Assistant

Engine: Trilingual Citizen Portal & Gemini RAG

Democratizes audit findings for citizens via the 5 Core Citizen Questions in English, Marathi, and Hindi. The AI Assistant answers public queries strictly with statutory page and observation citations.

`OUTPUT: Public Accountability & Cited Answers`

ENGINEERING SPECIFICATIONS • ZERO HALLUCINATION ARCHITECTURE

Full-Stack Architecture

Next.js 14 App Router & TypeScript

- React 18 Concurrent Features & Server Components
- Strict TypeScript typechecking (0 compilation errors across 26 App Router routes)
- Tailwind CSS enterprise styling with custom forest green palette (#0E2E23, #059669)
- Client-side state hydration via ClimateContext with multi-city support

Generative AI Integration

Google Gemini 1.5 Flash Vision & RAG

- Multi-modal policy document extraction parsing PDF tables and statutory text
- Strict structural isolation: AI extracts text; math is evaluated deterministically
- Retrieval-Augmented Citizen Assistant citing verified page & observation sources
- Trilingual civic synthesis in English, Marathi, and Hindi

Scientific Audit Engine

Deterministic Integrity Matrix v2.4

- Excel specification model parity: 60.7% aggregate portfolio fulfillment rate
- Excel gap thresholds: On Track (<=5%), At Risk (5-15%), Off Track (>15%)
- Calculated reporting overstatement math detecting discrepancies (+36.5%)
- Multi-source agreement logic evaluating satellite vs ground telemetry

Geospatial Visualization

Leaflet GIS & Public OpenStreetMap

- 100% Free OpenStreetMap & Esri satellite tiles requiring Zero API keys
- Interactive climatic layers: Monsoon rain radar (78 mm/hr) & heat islands (44.3°C)
- Ward-level observation pins tracking tree canopy, electric buses, and solar arrays
- Explicit scientific disclosure: SIMULATED DEMONSTRATION RECORD stamps

Cryptographic Provenance

Tamper-Evident SHA-256 State Ledger

- Standard 64-character SHA-256 hexadecimal state hashes for all formal audits
- Cryptographic audit trail linking timestamp, evaluated targets, and raw evidence
- Immutable verification ledger page logging historical outcomes chronologically
- Dynamic client/server SHA-256 digest generator for real-time audit verification

Automated Document Export

Native jsPDF & Binary Streaming

- Instantaneous client-side formal audit certificate generation
- Automated executive PDF walkthrough builder compiling high-resolution captures
- Dedicated binary streaming route (/api/download) delivering 18-page walkthrough
- Production print-ready layouts with vector icons and tabular matrices

GOVERNMENT DEPLOYMENT VALUE • FUTURE PRODUCTION HORIZONS

1. PRACTICAL MUNICIPAL VALUE (IMPLEMENTED)

Elimination of Departmental Greenwashing:

By cross-referencing BMC/BEST internal claims against Copernicus Sentinel-2 satellite indices, municipal commissioners immediately detect overstatements (e.g. +36.5% discrepancy in Ward K-East) before funds are disbursed to private contractors.

Actionable Engineering Decision Support:

The What-If Simulator allows urban transport planners to model fleet procurement against emissions abatement (INR 375 Cr for 200 e-buses + 3 corridors -> 17.8% reduction), transforming static policy reports into dynamic operational tools.

Democratized Public Accountability:

The Citizen Portal translates complex Earth observation data into plain language across English, Marathi, and Hindi, answering the 5 Core Citizen Questions and restoring civic trust in municipal governance.

Tamper-Evident SHA-256 Audit Trail:

Cryptographic state hashing prevents retroactive modification of environmental audit logs, providing international carbon registries and municipal green bond issuers with audited compliance proof.

2. PRODUCTION ROADMAP & FUTURE SCOPE

Live ESA Copernicus STAC API Pipeline:

Transition from pre-calibrated demonstration tiles to live automated satellite tasking via the ESA Copernicus Data Space Ecosystem API, processing automated Sentinel-2 Level-2A imagery on every 5-day orbital pass.

Real-Time SCADA IoT Waterway Sensors:

Direct telemetry ingestion from BMC ultrasonic water-level sensors along the Mithi river basin and arterial nullahs to provide real-time flash flood surge alarms to emergency management authorities.

State-Wide Urban Local Body Rollout:

Expansion of the platform from the Mumbai & Pune pilot jurisdictions to all 29 Municipal Corporations across Maharashtra under the Department of Environment and Climate Change (DoECC GoM).

Municipal Green Bond Verification Integration:

Integration with SEBI-regulated municipal green bond frameworks, linking coupon interest rates directly to satellite-verified carbon abatement performance metrics.